Matrix functions and stability

Temporal regularity of marginal matrix-product growth

Difficulty: challenging **Importance:** interesting to specialist

Status: Open

Rating rationale: Challenging because cancellations across coupled blocks resist general growth regularity; its immediate impact is on specialists in marginally stable matrix products.

Context and notation

The joint spectral radius of a nonempty compact set $\mathcal{M} \subset \mathbb{C}^{d\times d}$ is

$$\hat{\rho}(\mathcal{M}) = \lim_{k\to\infty} \max_{A_1,\dots,A_k\in\mathcal{M}} \|A_k \cdots A_1\|_2^{1/k}.$$

This definition also applies to finite real matrix sets. The ordinary spectral radius of one matrix is written $\rho(A)$.

For a compact nonempty $\mathcal{M} \subset \mathbb{R}^{d\times d}$ with $\hat{\rho}(\mathcal{M}) = 1$, define its maximal product norm at length k by

$$g_{\mathcal{M}}(k) = \max_{A_1,\dots,A_k\in\mathcal{M}} \|A_k \cdots A_1\|_2.$$

This problem concerns growth within one family over time. MF-07 instead requests a dimension-dependent bound uniform across families.

Problem statement

Does every family in this notation satisfy both

$$\exists c > 0\ \forall k,\ell \ge 1 : \quad g_{\mathcal{M}}(k+\ell) \ge c\, g_{\mathcal{M}}(k)$$

and

$$\forall \ell \ge 1\ \exists C_\ell > 0\ \forall k \ge 1 : \quad g_{\mathcal{M}}(\ell k) \le C_\ell\, g_{\mathcal{M}}(k)?$$

Constants may depend on $\mathcal{M}$.

Reference and status evidence

Varney and Morris, On marginal growth rates of matrix products, Linear Algebra Appl. 709 (2025), 132–163, DOI, §4, Definition 4.1 and §7, Question 1. The source names these properties weak increase and weak upper regular variation. Searches for both property names, the authors, and subsequent matrix-product growth papers located no resolution.

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Rechecked Varney–Morris, §7, Question 1. Both weak increase and weak upper regular variation remain posed; the source explains why cancellation defeats its current argument. Property-name and author follow-up searches found no resolution of the two-part target.